Capstone Three Proposal: Consumer Complaint Routing and Relief Prediction

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Problem statement

Banks, lenders and credit bureaus receive large numbers of consumer complaints. Each one has to be sent to the right team, and someone has to decide which to review first. This project uses the text of a complaint to predict two things: its product and issue category, so that it can be routed, and whether the company closed it with monetary relief rather than with an explanation. A compliance team could use the second prediction to look first at the complaints most likely to involve real financial harm. The project also tests a question that matters for any tool like this: does the narrative add information beyond what the complaint form already records, such as the product, the company, the state and how the complaint was submitted?

Problem identification

Context

The Consumer Financial Protection Bureau (CFPB) collects complaints about financial products and publishes them together with the company's response. Most closed complaints end with an explanation and no relief. An AARP review of the database put monetary relief at 7 to 9 percent of complaints in each year it examined. Credit reporting is the largest single category, so volume is concentrated in a few products. The first notebook will measure the actual rate in the narrative subset.

Criteria for success

- Routing: macro-averaged F1 for product classification clearly above a majority-class baseline, with per-class results reported so that small products are not hidden by large ones.
- Relief: average precision (area under the precision-recall curve) above the base rate and above a model that uses only the structured fields, measured on complaints from a later period than the training data. Also report precision among the top 10 percent of complaints ranked by predicted relief, which is how a review queue would use the model.
- If the narrative does not improve on the structured-field model, report that result and explain it.
- A documented, reproducible pipeline in a GitHub repository that a mentor can run.

Scope

In scope are the complaints in the snapshot that include a narrative, and two supervised tasks: multi-class product and issue classification, and binary relief prediction. Models: TF-IDF with logistic regression, gradient boosting on the structured fields, and pretrained sentence embeddings if time allows. The relief task compares "closed with monetary relief" against "closed with explanation". Complaints closed with non-monetary relief or still in progress are left out, with a sensitivity check that counts non-monetary relief as a negative. Out of scope are deployment, fine-tuning large language models, scoring individual companies, and causal claims about why relief was granted.

Constraints

- Rare outcome: monetary relief is a small minority, so the project uses average precision, class weights and threshold analysis instead of accuracy.
- Product mix: credit reporting dominates the data and seldom ends in monetary relief, so a model can score well just by recognizing the product. Results are compared with a structured-field baseline and reported within the largest products.
- Label meaning: the response is recorded by the company, so the model predicts what the company recorded, not whether the consumer was harmed.
- Selection: narratives exist only for consumers who opted in to publication, so results describe that group.
- Frozen data: on August 14, 2026 the CFPB stopped publishing complaint narratives. The snapshot is a fixed historical extract, which suits a capstone, but a model built on it cannot be refreshed with new narratives. The final report will say so.
- Time drift: relief rates change over the years, so data are split by complaint date, not at random.
- Size: the snapshot has about 3.1 million rows (about 2 GB), so modeling uses the narrative subset and, for the slowest models, a sample.

Stakeholders

- Primary: complaint-handling and compliance teams at banks, lenders and credit bureaus.
- Secondary: consumer-protection researchers and advocates who study how complaints are resolved.
- Springboard mentor and reviewers, who assess the work for rigor.

Key data source

Source	CFPB Consumer Complaint Database, as the Hugging Face snapshot (consumer-finance-complaints)
Size	3,079,747 complaints; narratives only for consumers who opted in
Key fields	Product, issue, complaint narrative, company, state, submitted via, company response to consumer, timely response
Targets	Product and issue category (routing); monetary relief versus explanation (priority)
License	CFPB public data, CC0

Proposed approach

1. Wrangling and exploratory analysis: count narratives, clean the text, and measure relief rates by product, company and year.
2. Baselines: majority class, and models on the structured fields alone.
3. Text models: TF-IDF with logistic regression, then sentence embeddings with a linear or boosted model.
4. Evaluation: time-based split, macro F1, average precision, top-decile precision, and error analysis of relief cases the model misses.
5. Output: a final report, slides, and the notebooks in a GitHub repository.